Chapter 3 - Data Transformation

2026-03-19

```r
library(nycflights13)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Review the data

1. **glimpse(flights)**

2. **view(flights)**

3. **print(flights)**

dplyr Basics

1. **pip |> or %>%**

```r
flights %>%
  filter(dest == 'IAH') %>%
  group_by(year, month, day) %>%
  summarize(
    arr_delya = mean(arr_delay, na.rm = TRUE)
  )
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by year, month, and day.
## i Output is grouped by year and month.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(year, month, day))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
## # A tibble: 365 x 4
## # Groups:   year, month [12]
##     year month   day arr_delya
##    <int> <int> <int>     <dbl>
##  1  2013     1     1     17.8
##  2  2013     1     2      7
##  3  2013     1     3     18.3
##  4  2013     1     4     -3.2
##  5  2013     1     5     20.2
##  6  2013     1     6      9.28
##  7  2013     1     7     -7.74
##  8  2013     1     8      7.79
##  9  2013     1     9     18.1
## 10  2013     1    10      6.68
## # i 355 more rows
```

Rows - Most Important verbs:

1. filters(): keeps row based on the values of the columns

- =; !=; >; >=;<;<=
- %in% c(): check if in a list

2. arrange(): changes the order of the rows based on the value of the columns

- desc()

3. distinct(): finds all the unique rows in a database.

- .keep_all = TRUE

4. count()

- sort = TRUE

5. head(10)

- only show 10 rows

6. nrow(): show how many rows available

flights %>%

nrow()

7. slice_max(distance, n = 5)

distinct(distance) %>% # select distinct first

slice_min(distance, n = 5) # get the first 5 mins

8. select()

- list of the columns

1. Filters()

- =; !=; >; >=;<;<=
- %in% c(): check if in a list

```r
# filter flight depart delay more than 120
# <, <=, >, >=, !=
flights %>%
  filter(dep_delay > 120)
```

```
## # A tibble: 9,723 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      848           1835       853     1001           1950
##  2  2013     1     1      957            733       144     1056            853
##  3  2013     1     1     1114            900       134     1447           1222
##  4  2013     1     1     1540           1338       122     2020           1825
##  5  2013     1     1     1815           1325       290     2120           1542
##  6  2013     1     1     1842           1422       260     1958           1535
##  7  2013     1     1     1856           1645       131     2212           2005
##  8  2013     1     1     1934           1725       129     2126           1855
##  9  2013     1     1     1938           1703       155     2109           1823
## 10  2013     1     1     1942           1705       157     2124           1830
## # i 9,713 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```r
# &, |
flights %>%
  filter(month == 1 & day == 1)
```

```
## # A tibble: 842 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      517            515         2      830            819
##  2  2013     1     1      533            529         4      850            830
##  3  2013     1     1      542            540         2      923            850
##  4  2013     1     1      544            545        -1     1004           1022
##  5  2013     1     1      554            600        -6      812            837
##  6  2013     1     1      554            558        -4      740            728
##  7  2013     1     1      555            600        -5      913            854
##  8  2013     1     1      557            600        -3      709            723
##  9  2013     1     1      557            600        -3      838            846
## 10  2013     1     1      558            600        -2      753            745
## # i 832 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```r
# %in%: keeps rows where the variable equals one of the vlaues on the right.
flights %>%
  filter(month %in% c(1, 2))
```

```
## # A tibble: 51,955 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      517            515         2      830            819
##  2  2013     1     1      533            529         4      850            830
##  3  2013     1     1      542            540         2      923            850
##  4  2013     1     1      544            545        -1     1004           1022
##  5  2013     1     1      554            600        -6      812            837
##  6  2013     1     1      554            558        -4      740            728
##  7  2013     1     1      555            600        -5      913            854
##  8  2013     1     1      557            600        -3      709            723
##  9  2013     1     1      557            600        -3      838            846
## 10  2013     1     1      558            600        -2      753            745
## # i 51,945 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```r
# asign to a new variable
jan1 <- flights %>%
  filter(month == 1 & day == 1)
```

2. arrange()

- desc(): sort by columns

```r
# desc(): decrease order
flights %>%
  arrange(year, month, day, desc(dep_delay))
```

```
## # A tibble: 336,776 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      848           1835       853     1001           1950
##  2  2013     1     1     2343           1724       379      314           1938
##  3  2013     1     1     1815           1325       290     2120           1542
##  4  2013     1     1     2205           1720       285       46           2040
##  5  2013     1     1     1842           1422       260     1958           1535
##  6  2013     1     1     2115           1700       255     2330           1920
##  7  2013     1     1     2006           1630       216     2230           1848
##  8  2013     1     1     2312           2000       192       21           2110
##  9  2013     1     1     1942           1705       157     2124           1830
## 10  2013     1     1     1938           1703       155     2109           1823
## # i 336,766 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

3. distinct()

- .keep_all = TRUR: keep the other columns when filtering for unique rows

```r
flights %>%
  distinct(month)
```

```
## # A tibble: 12 x 1
##    month
##    <int>
##  1     1
##  2    10
##  3    11
##  4    12
##  5     2
##  6     3
##  7     4
##  8     5
##  9     6
## 10     7
## 11     8
## 12     9
```

```r
flights %>%
  distinct(origin, dest)
```

```
## # A tibble: 224 x 2
##    origin dest
##    <chr>  <chr>
##  1 EWR    IAH
##  2 LGA    IAH
##  3 JFK    MIA
##  4 JFK    BQN
##  5 LGA    ATL
##  6 EWR    ORD
##  7 EWR    FLL
##  8 LGA    IAD
##  9 JFK    MCO
## 10 LGA    ORD
## # i 214 more rows
```

```r
flights %>%
  distinct(origin, dest, .keep_all = TRUE)
```

```
## # A tibble: 224 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      517            515         2      830            819
##  2  2013     1     1      533            529         4      850            830
##  3  2013     1     1      542            540         2      923            850
##  4  2013     1     1      544            545        -1     1004           1022
##  5  2013     1     1      554            600        -6      812            837
```

```
## 6 2013 1 1 554 558 -4 740 728
## 7 2013 1 1 555 600 -5 913 854
## 8 2013 1 1 557 600 -3 709 723
## 9 2013 1 1 557 600 -3 838 846
## 10 2013 1 1 558 600 -2 753 745
## # i 214 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

4. count()

- sort = TRUE

```r
flights %>%
  count(origin, dest, sort = TRUE)
```

```
## # A tibble: 224 x 3
##    origin dest      n
##    <chr>  <chr> <int>
##  1 JFK    LAX   11262
##  2 LGA    ATL   10263
##  3 LGA    ORD    8857
##  4 JFK    SFO    8204
##  5 LGA    CLT    6168
##  6 EWR    ORD    6100
##  7 JFK    BOS    5898
##  8 LGA    MIA    5781
##  9 JFK    MCO    5464
## 10 EWR    BOS    5327
## # i 214 more rows
```

Exercises

```r
names(flights)
```

```
##  [1] "year"           "month"          "day"            "dep_time"
##  [5] "sched_dep_time" "dep_delay"      "arr_time"       "sched_arr_time"
##  [9] "arr_delay"      "carrier"        "flight"         "tailnum"
## [13] "origin"         "dest"           "air_time"       "distance"
## [17] "hour"           "minute"         "time_hour"
```

- Had an arrival delay of two or more hours

```r
flights %>%
  filter(arr_delay > 120)
```

```
## # A tibble: 10,034 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
```

```
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      811            630       101     1047            830
##  2  2013     1     1      848           1835       853     1001           1950
##  3  2013     1     1      957            733       144     1056            853
##  4  2013     1     1     1114            900       134     1447           1222
##  5  2013     1     1     1505           1310       115     1638           1431
##  6  2013     1     1     1525           1340       105     1831           1626
##  7  2013     1     1     1549           1445        64     1912           1656
##  8  2013     1     1     1558           1359       119     1718           1515
##  9  2013     1     1     1732           1630        62     2028           1825
## 10  2013     1     1     1803           1620       103     2008           1750
## # i 10,024 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

- Flew to Houston (`IAH` or `HOU`)

```
flights %>%
  filter(dest %in% c('IAH','HOU'))
```

```
## # A tibble: 9,313 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      517            515         2      830            819
##  2  2013     1     1      533            529         4      850            830
##  3  2013     1     1      623            627        -4      933            932
##  4  2013     1     1      728            732        -4     1041           1038
##  5  2013     1     1      739            739         0     1104           1038
##  6  2013     1     1      908            908         0     1228           1219
##  7  2013     1     1     1028           1026         2     1350           1339
##  8  2013     1     1     1044           1045        -1     1352           1351
##  9  2013     1     1     1114            900       134     1447           1222
## 10  2013     1     1     1205           1200         5     1503           1505
## # i 9,303 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

- Were operated by United, American, or Delta

```
flights %>%
  filter(carrier %in% c('UA', 'AA', 'DA'))
```

```
## # A tibble: 91,394 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      517            515         2      830            819
##  2  2013     1     1      533            529         4      850            830
##  3  2013     1     1      542            540         2      923            850
##  4  2013     1     1      554            558        -4      740            728
##  5  2013     1     1      558            600        -2      753            745
```

```
## 6  2013     1     1      558            600        -2      924            917
## 7  2013     1     1      558            600        -2      923            937
## 8  2013     1     1      559            600        -1      941            910
## 9  2013     1     1      559            600        -1      854            902
## 10 2013     1     1      606            610        -4      858            910
## # i 91,384 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```r
flights %>%
  filter(month %in% c(7, 8, 9))
```

```
## # A tibble: 86,326 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
## 1  2013     7     1        1           2029       212      236           2359
## 2  2013     7     1        2           2359         3      344            344
## 3  2013     7     1       29           2245       104      151              1
## 4  2013     7     1       43           2130       193      322             14
## 5  2013     7     1       44           2150       174      300            100
## 6  2013     7     1       46           2051       235      304           2358
## 7  2013     7     1       48           2001       287      308           2305
## 8  2013     7     1       58           2155       183      335             43
## 9  2013     7     1      100           2146       194      327             30
## 10 2013     7     1      100           2245       135      337            135
## # i 86,316 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```r
flights %>%
  filter(arr_delay >= 120 & dep_delay <= 0)
```

```
## # A tibble: 29 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
## 1  2013     1    27     1419           1420        -1     1754           1550
## 2  2013    10     7     1350           1350         0     1736           1526
## 3  2013    10     7     1357           1359        -2     1858           1654
## 4  2013    10    16      657            700        -3     1258           1056
## 5  2013    11     1      658            700        -2     1329           1015
## 6  2013     3    18     1844           1847        -3       39           2219
## 7  2013     4    17     1635           1640        -5     2049           1845
## 8  2013     4    18      558            600        -2     1149            850
## 9  2013     4    18      655            700        -5     1213            950
## 10 2013     5    22     1827           1830        -3     2217           2010
## # i 19 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```r
flights %>%
  filter(dep_delay >= 50 & dep_delay - arr_delay <= 30)
```

```
## # A tibble: 30,217 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      811            630       101     1047            830
##  2  2013     1     1      826            715        71     1136           1045
##  3  2013     1     1      848           1835       853     1001           1950
##  4  2013     1     1      957            733       144     1056            853
##  5  2013     1     1     1114            900       134     1447           1222
##  6  2013     1     1     1120            944        96     1331           1213
##  7  2013     1     1     1255           1200        55     1451           1330
##  8  2013     1     1     1301           1150        71     1518           1345
##  9  2013     1     1     1337           1220        77     1649           1531
## 10  2013     1     1     1356           1259        57     1538           1438
## # i 30,207 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

1. **Sort `flights` to find the flights with the longest departure delays. Find the flights that left earliest in the morning.**

```r
names(flights)
```

```
##  [1] "year"           "month"          "day"            "dep_time"
##  [5] "sched_dep_time" "dep_delay"      "arr_time"       "sched_arr_time"
##  [9] "arr_delay"      "carrier"        "flight"         "tailnum"
## [13] "origin"         "dest"           "air_time"       "distance"
## [17] "hour"           "minute"         "time_hour"
```

```r
flights %>%
  arrange(desc(dep_delay))
```

```
## # A tibble: 336,776 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     9      641            900      1301     1242           1530
##  2  2013     6    15     1432           1935      1137     1607           2120
##  3  2013     1    10     1121           1635      1126     1239           1810
##  4  2013     9    20     1139           1845      1014     1457           2210
##  5  2013     7    22      845           1600      1005     1044           1815
##  6  2013     4    10     1100           1900       960     1342           2211
##  7  2013     3    17     2321            810       911      135           1020
##  8  2013     6    27      959           1900       899     1236           2226
##  9  2013     7    22     2257            759       898      121           1026
## 10  2013    12     5      756           1700       896     1058           2020
## # i 336,766 more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```r
flights %>%
  arrange(dep_time) %>%
  select(year, month, day, carrier, flight, dep_delay) %>%
  head(10)
```

```
## # A tibble: 10 x 6
##     year month   day carrier flight dep_delay
##    <int> <int> <int> <chr>    <int>     <dbl>
##  1  2013     1    13 B6          22        72
##  2  2013     1    31 WN         530       181
##  3  2013    11    13 B6        1503         2
##  4  2013    12    16 B6         839         2
##  5  2013    12    20 B6        1503         2
##  6  2013    12    26 B6        1503         2
##  7  2013    12    30 B6         839         2
##  8  2013     2    11 WN         530       181
##  9  2013     2    24 B6         608        76
## 10  2013     3     8 B6         739         6
```

1. **Sort `flights` to find the fastest flights. (Hint: Try including a math calculation inside of your function.)**

```r
flights %>%
  mutate(miles_per_hour = distance / (air_time/60)) %>%
  select(year, month, day, carrier, flight, distance, air_time, miles_per_hour) %>%
  arrange(desc(miles_per_hour))
```

```
## # A tibble: 336,776 x 8
##     year month   day carrier flight distance air_time miles_per_hour
##    <int> <int> <int> <chr>    <int>    <dbl>    <dbl>          <dbl>
##  1  2013     5    25 DL        1499      762       65           703.
##  2  2013     7     2 EV        4667     1008       93           650.
##  3  2013     5    13 EV        4292      594       55           648
##  4  2013     3    23 EV        3805      748       70           641.
##  5  2013     1    12 DL        1902     1035      105           591.
##  6  2013    11    17 DL         315     1598      170           564
##  7  2013     2    21 B6         707     1598      172           557.
##  8  2013    11    17 AA         936     1623      175           556.
##  9  2013    11    16 DL         347     1598      173           554.
## 10  2013    11    16 B6        1503     1598      173           554.
## # i 336,766 more rows
```

1. **Was there a flight on every day of 2013?**

```r
flights %>%
  distinct(year, month, day) %>%
  nrow()
```

```
## [1] 365
```

1. **Which flights traveled the farthest distance? Which traveled the least distance?**

```r
flights %>%
  arrange(desc(distance)) %>%
  head(1)
```

```
## # A tibble: 1 x 19
##    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
## 1  2013     1     1      857            900        -3     1516           1530
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```r
flights %>%
  distinct(distance) %>%
  slice_max(distance, n = 5)
```

```
## # A tibble: 5 x 1
##   distance
##      <dbl>
## 1     4983
## 2     4963
## 3     3370
## 4     2586
## 5     2576
```

```r
flights %>%
  distinct(distance) %>%
  slice_min(distance, n=5)
```

```
## # A tibble: 5 x 1
##   distance
##      <dbl>
## 1       17
## 2       80
## 3       94
## 4       96
## 5      116
```

1. Does it matter what order you used `filter()` and `arrange()` if you're using both? Why/why not? Think about the results and how much work the functions would have to do.

   - the result would be the same
   - the filter() -> arrange() would be more efficient than arrange() -> filter(). filter() -> arrange() doesn't need to the arrange() the whole data set, only the ones after the filter()